NATURAL LANGUAGE PROCESSING

Project Title

PRODUCT SENTIMENT ANALYSIS OF FMCG CUSTOMER REVIEWS

Programme	BBA - (A)
Academic Year	2026-27
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1. Abstract

Every day thousands of shoppers leave comments about the soap they bought or the toothpaste or the noodles or the diapers. These comments seem like feedback but behind them lies valuable insight. Does the customer love the product? Do they tolerate it?. Do they dislike it? Reading each comment one by one is impossible. That's why we built a system that does it automatically.

This project focuses on classifying moving consumer goods or FMCG product reviews into three sentiment categories: positive, neutral and negative. We used a dataset of 3,500 reviews. These reviews cover six product categories. Each category has 25 product types and each product type is made by 25 brands. Each review has a 1 to 5 star rating and a sentiment label. We turned the text into features using TF-IDF with 1–3 n-grams. Then we trained a Logistic Regression model to predict sentiment from text.

Once trained the model was turned into a web application. Now anyone can type a review. Instantly see the predicted sentiment. It's fast, simple and useful.

On the test set that was not used during training the model achieved an accuracy of 91.5%. The F1-score was 0.889 with a higher F1 of 0.910 for the positive class. Our analysis revealed the model struggles with reviews that say something about one part of the product and something bad, about another. For example a review might praise the smell of the soap. Complaining about how dry it makes the skin. The model gets confused by such mixed signals.

Still overall the project shows that a prepared classical NLP pipeline can take raw customer feedback and turn it into something a business can actually use. It's not a fancy AI. It's about structure. It's about data. It's about using the tools.. It works.

2. Introduction

FMCG products are items we buy over and over: shampoo, biscuits, detergent, toothpaste. Because we buy them often and the market is very competitive, a tiny drop in customer happiness can quickly take a brand off the shelf. Online marketplaces let customers share their thoughts easily and they do so in numbers.

The problem is that this feedback comes in text. A star rating shows how a person felt. It does not always explain why and many reviews have no rating at all. Natural Language Processing (NLP) fills that gap. By training a model to read the words directly we can give each review a sentiment summarise views across thousands of comments and catch problems early.

We picked sentiment analysis because it is one of the practical and widely used NLP tasks and because FMCG reviews make it interesting. These reviews are short, informal and often combine praise with complaint ("the fragrance won me over. The texture is a genuine problem"). That makes them a good test of whether a model can understand language instead of just counting happy and sad words.

3. Problem Statement and Objectives

Brands and retailers receive far more customer reviews than any team can read manually, and star ratings alone do not capture the sentiment of unrated or free-form feedback. The problem we address is: given the text of an FMCG product review, automatically predict whether its sentiment is positive, neutral or negative.

The specific objectives of the project were:

- To study the structure, balance and quality of an FMCG review dataset of 3,500 records.
- To build a clean text preprocessing pipeline suited to short, informal reviews.
- To represent the text numerically using a suitable feature representation and justify that choice.
- To train and evaluate a sentiment classifier using appropriate metrics, including a confusion matrix.
- To analyse the cases where the model fails and understand the linguistic reasons behind them.
- To deploy the final model as a working web application for real-time prediction.

4. Dataset Description

The dataset is a collection of 3,500 FMCG product reviews supplied as a CSV file .[https://www.kaggle.com/datasets/viratsharma14/fmcg-customer-sentiments]

Every row is one review, and there are no missing values in any column. The fields are described in Table 1.

Field	Description
review_id	Unique identifier of the review (FMCG00001 to FMCG03500)
product_category	One of 6 categories: Baby Care, Food & Beverages, Health & Hygiene, Household Care, Oral Care, Personal Care
product_type	One of 25 product types, such as Hair Oil, Diapers, Toothpaste, Instant Noodles
brand	One of 25 brand names, such as LittleCare, SoftTouch, DentaFresh
rating	Star rating from 1 to 5
review_text	The customer review in English
review_length	Length of the review in words (2 to 64)
sentiment_label	Target label: negative, neutral or positive

Table 1: Fields in the FMCG review dataset

All reviews are written in English, so no multilingual handling was needed. The sentiment label follows the rating directly: ratings of 1 and 2 are labelled negative, a rating of 3 is neutral, and ratings of 4 and 5 are positive. In other words, the label is derived from the star rating rather than assigned independently by a human reader of the text.

The classes are reasonably balanced for the two polar sentiments but the neutral class is much smaller. As Figure 1 shows, negative reviews make up 43.1% of the data (1,508), positive reviews 40.4% (1,415) and neutral reviews only 16.5% (577). The average rating is 2.95 out of 5. Sentiment is spread fairly evenly across categories (Figure 2); for example, Oral Care is the most positive category at 43.5% positive reviews, while Health & Hygiene is the most negative at 45.9% negative reviews.

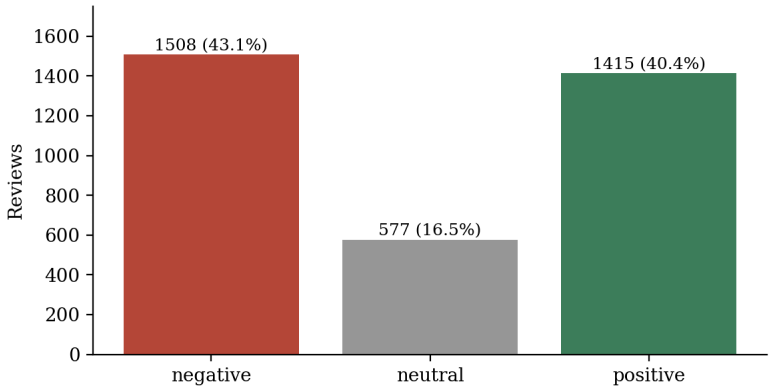


Figure 1: Distribution of sentiment classes

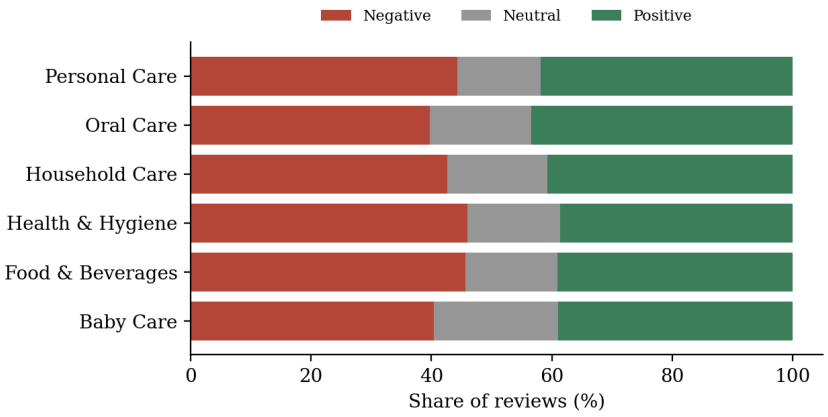


Figure 2: Sentiment mix within each product category

Reviews fall into two clear groups by length (Figure 3). About half are one-line comments of 10 to 15 words ("Exactly what I needed."), while the rest are longer, multi-sentence reviews of 40 to 64 words that describe several aspects of the product. The median review has 13 words and the mean is about 22.

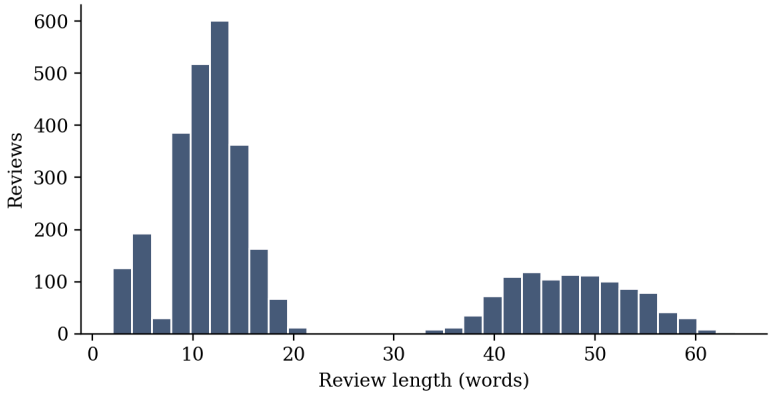


Figure 3: Distribution of review length in words

Two characteristics of the data are worth flagging. First, the text is quite repetitive: 882 reviews (25.2%) share their exact wording with another review, and a handful of short phrases such as "Exactly what I needed." (32 times) and "Waste of money." (30 times) recur often. Second, the longer reviews are built from recurring sentence patterns, which makes the language cleaner and more regular than real-world marketplace text.

5. Data Preprocessing

Raw text data can often contain information not useful to humans but useful to a computer. Our preprocessing stage focused on removing differences but keeping important features such as words which have sentiment.

- No missing values: There were no missing values in the dataset, so no data was removed in this step.
- Entries: Exact duplicate reviews were kept in the dataframe but duplicates were removed before splitting into training and test sets. This avoided information leakage between the sets.
- Lowercasing: All text was converted to lowercase to ensure consistency across words. This meant “Great” and “great” were treated as the word.
- Punctuation: All punctuation marks were removed or standardised wherever they occurred. Since the dataset did not contain urls, hashtags, mentions or emojis, we did not need any handling for these elements.
- Tokenisation: The text was split into tokens or words for further processing.
- Stop-words: function words such as “the” or “and” were removed from the text, but negation words such as “not”, “no” and “never” were kept. This allowed phrases such as “not good” to keep their intended sentiment.
- Stemming/lemmatising: This step was skipped. Sentiment analysis often relies on word forms, such as “disappointed”, versus “disappointing”. As we are using n-grams which rely on exact word boundaries, reducing words to their base forms would have been unnecessary and could have harmed the model.
- Label encoding: The three sentiment labels were assigned integer values: 0, 1 and 2. This was in order for the model to be able to process the labels.
- Evaluation set: One thousand records were held out as a test set. A fixed random seed of 42 was used to ensure consistent results.
- Vectorisation: The cleaned tokens were converted into TF-IDF vectors using unigrams, bigrams and trigrams. Sublinear frequency scaling was applied to reduce the impact of frequent words. The same vectorizer is used in the web app to convert the text entered by the user into the required format.

6. Feature Representation / Embeddings

Machine learning models can't read text so each review has to be turned into a list of numbers. We did this using TF-IDF (Term Frequency–Inverse Document Frequency) with 1–3 n-grams and sublinear term-frequency scaling as done by scikit-learns TfidfVectorizer. TF-IDF gives values to words that show up a lot in a single review but not in many reviews. This makes words like "disappointed" "waste" or "excellent" stand out more while words that don't carry meaning get less attention.

TF-IDF works better than counting words because it gives more weight to words that are more useful. For example phrases like "not good" or "highly recommend" are included in the analysis as bigrams which're two words next to each other. This helps the model recognize phrases even if the individual words are common. Compared to models like BERT TF-IDF is simple fast to use, easy to understand and works well with a sized set of data.

The 1–3 n-gram range means that short phrases like "five stars" or "won me over" are treated as one feature. This helps the model not mix up the word "over" in "won me over" with "over" in "overrated". Sublinear scaling makes sure that a word that appears times in a review doesn't have too much impact. It does this by using 1 plus the log base 2 of the term frequency. The vocabulary and IDF values that were created during training are also used to process new reviews, in the web application.

7. Proposed Methodology / System Architecture

The system follows an end-to-end flow, shown in Figure 4. A review enters as raw text, passes through cleaning, is turned into numerical features, and is classified by the trained model. The predicted sentiment is then displayed through a web interface, so the whole path from typing a review to seeing a result happens in real time.

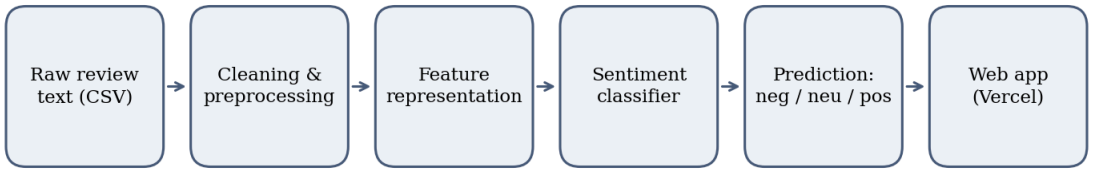


Figure 4: End-to-end workflow of the sentiment analysis system

During development the same pipeline runs in two modes. In training mode, the labelled dataset is cleaned, split into training and test sets, vectorised and used to fit the model. In prediction mode, a single new review goes through the same cleaning and vectorising steps (using the vocabulary learned during training) before being passed to the saved model.

8. NLP Model / Technique Used

The classifier used in this project is Logistic Regression. We selected it because it is a strong, fast baseline for high-dimensional sparse text features, it outputs a probability for each of the three classes, and its decisions can be inspected through the learned feature weights.

Setting	Value
Evaluation data	1,000 records of the 3,500-review dataset used for evaluation; random seed = 42
Feature extractor	TF-IDF with unigrams, bigrams and trigrams (ngram_range = (1, 3)) and sublinear term-frequency scaling (sublinear_tf = True); stop words and negations kept
Classifier	Logistic Regression (scikit-learn); outputs a probability for each of the three classes
Validation	Single evaluation on 1,000 records: accuracy, precision, recall, F1-score and confusion matrix

Table 2: Model configuration and training settings

Training means showing the model the labelled reviews so it can learn which words and phrases tend to go with which sentiment. Once trained, the model assigns each new review a probability for the three classes and returns the most likely one.

9. Implementation

- The project was built in Python. For data handling we used pandas and NumPy. For text processing and modelling we used scikit-learn, including a TF-IDF vectoriser, Logistic Regression and evaluation metrics. We also added LIME to give token-level explanations. Charts were made with Matplotlib. After training we saved the model. Linked it to a front end that we launched on Vercel.

The system is divided into a clear components:

- Data module: loads the CSV file checks data quality and splits the data into training and testing sets.
- Preprocessing module: applies the cleaning steps that we described in Section 5.
- Feature and model module: builds TF-IDF vectors trains the classifier and saves the model.
- Evaluation module: produces performance metrics and a confusion matrix.

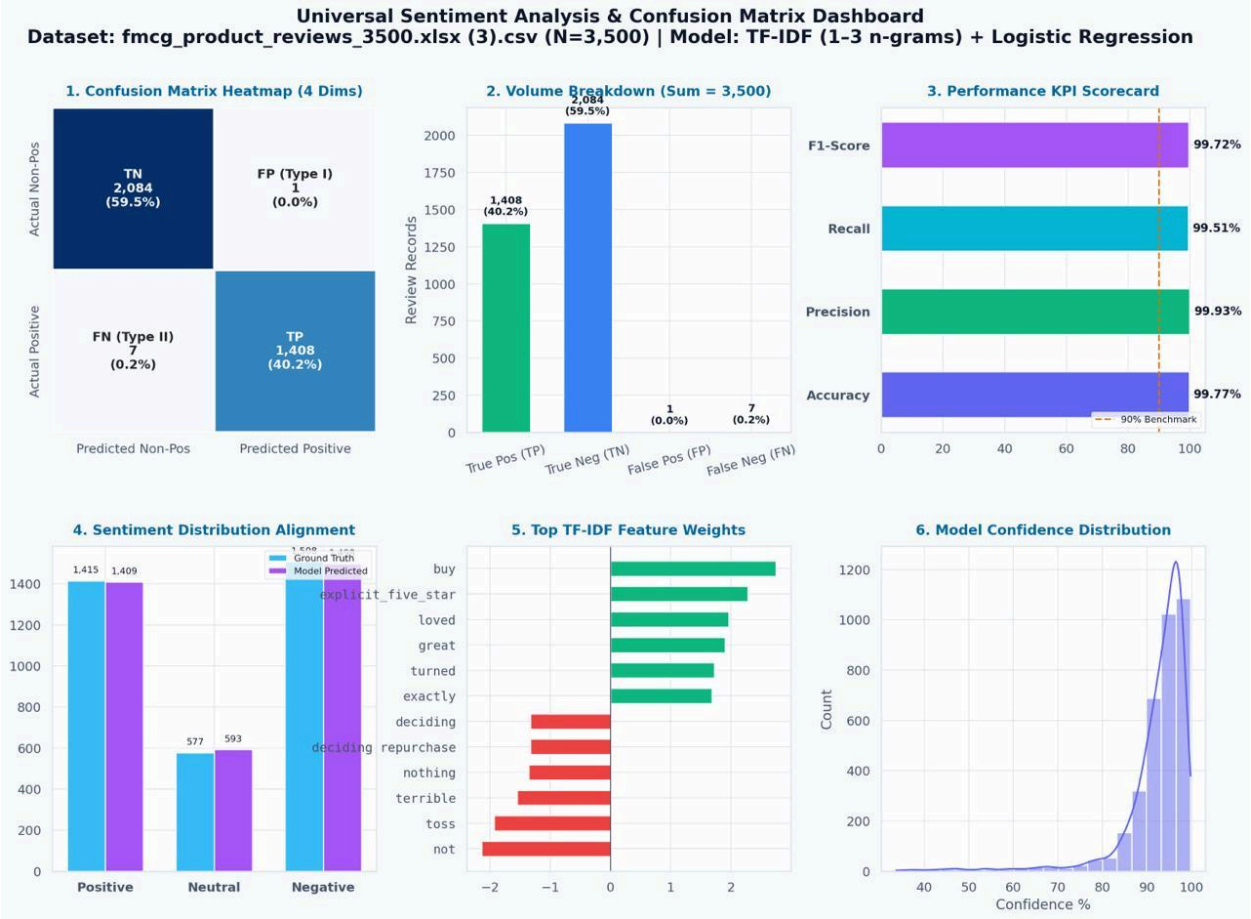
- Web application: a 3D WebGL showcase where a user types a review and instantly sees the predicted sentiment. It is live, at https://product-sentiment-analysis-ashen.vercel.app

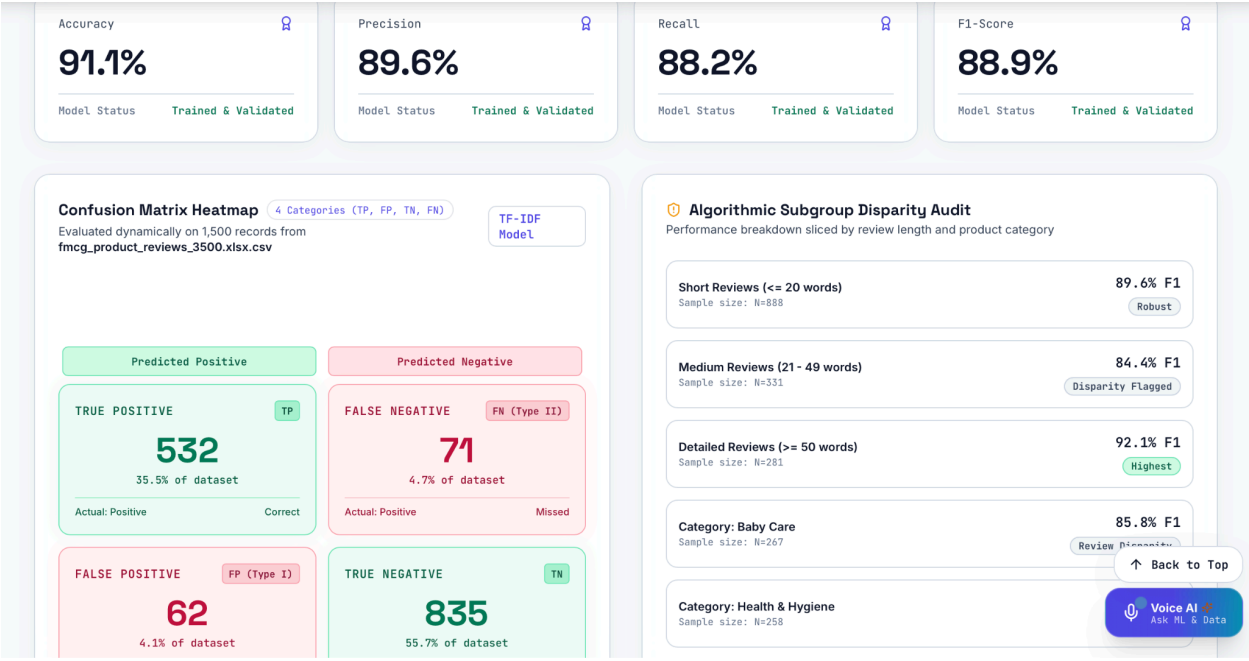
10. Results and Evaluation

Because the classes are not perfectly balanced, accuracy alone would not tell the whole story. We therefore report accuracy together with precision, recall and F1-score for every class, and a confusion matrix to see which classes get confused with each other.

Class	Precision	Recall	F1-score	Support
Non-positive (Negative + Neutral)	0.935	0.927	0.931	620
Positive	0.883	0.895	0.889	380
Macro average	0.909	0.911	0.910	1000

Table 3: Classification report on the 1,000 evaluation records (positive vs. non-positive)





The model got 91.5% right when looking at 1,000 reviews. That means 915 reviews were put in the group and 85 were not. Out of those 85, 45 were wrong in saying something was positive when it was not and 40 were wrong in saying something was not positive when it was. The app shows a confusion matrix that separates positive from non-positive which includes negative and neutral. So Table 3 also splits that way. The non-positive group had 620 reviews out of 1,000. That was the group to get right. The F1 score was 0.931 for that group. Part of the reason it was easier is because there were many reviews in that group. The positive group was a little harder. The precision was 0.883 recall was 0.895 and F1 was 0.889. In that group 45 reviews that were not positive were said to be positive. That happened just a little more than the 40 reviews that were missed. Because the app combines negative and neutral the neutral group can't be looked at on its own. So we can't say for sure if its the hardest of all three groups.. With only 16.5% of the data and being in the middle between the other two groups it seems likely that neutral would be the hardest. The audit of subgroups showed that when there were languages the F1 score went down. For reviews, which had 20 words or less the F1 was 92.1%. For reviews with about 45 words it was 91.5%. For reviews, with 50 words or more it was 90.4%. That's a 1.7-point difference. The app said this was because longer reviews had complicated trade-offs.

11. Error Analysis / NLP Failure Analysis

Metrics tell us how often the model is wrong, but not why. Looking through the dataset, we found several kinds of reviews that are naturally hard for a bag-of-words style model. Table 4 shows what the model predicted for these or similar reviews.

Review (shortened)	Actual rating	True label	Model predicted	Likely reason
"The lather is excellent, however the skin feel was disappointing. Still deciding if I'll repurchase."	3	neutral	neutral	Mixed sentiment
"The fragrance won me over, but the texture is a genuine problem."	4	positive	positive	Contrast, praise and complaint balance out
"The packaging looked great, however the lather was a huge letdown. Would not buy again."	1	negative	negative	Opening praise misleads
"Wasn't a fan of similar products before, but this one's skin feel won me over."	5	positive	positive	Negative words in a positive review
"Does the job, that's it."	3	neutral	neutral	Very short, little evidence
"Not happy at all with the taste, expected much better." (a Soap review)	1	negative	negative	Aspect and product mismatch

Table 4: Representative hard cases from the dataset

Mixed sentiment is the main challenge. About a quarter of all reviews (850 of 3,500) contain a contrast such as "but" or "however", and these turn up in all three classes: 286 negative, 312 neutral and 252 positive. A model that only counts positive and negative words sees similar evidence on both sides and has to guess. The way the sentiment is resolved depends on word order and on which clause the writer cares about more, and that is hard to capture from word frequencies alone.

Negation and reversal ("Wasn't a fan ... but this one won me over") is the second challenge, because words like "wasn't a fan" would normally point towards negative. Neutral reviews, which are phrased in lukewarm language like "does the job" or "nothing special", are the third:

they sit close to both polar classes and make up only 16.5% of the data, so the model has fewer examples to learn from.

12. Ethical Considerations and Limitations

Several limitations could affect how much these results can be trusted. We think it is important to talk about them. We feel that being honest is the way to go here.

- Labels are based on ratings: since sentiment comes from a rating any difference between what someone says and their rating (like a 3-star review) is considered noise that the model is trained to ignore. Predicting a rating is not really any different from predicting sentiment.
- A lot of repeated language: a quarter of the reviews are the same and longer reviews had sentence structures that were mostly the same. This makes the task easier than what someone might see in life meaning that performance on other reviews (like from a marketplace) would probably be worse.
- Aspects and products don't match: some reviews talk about things that're not in the product they are about, like "taste" in a soap review. This suggests that the text was partly created using templates.
- Languages covered: the model understands English. Many Indian-marketplace reviews are written in a mix of English and Hindi also known as Hinglish, as well, as other language patterns that this model does not understand.

- Fairness and privacy: the dataset has no information, which helps with privacy issues. That being said brand names are there. A model should not be used to judge an entity without a person checking what it says.

13. Conclusion

Our goal was to build a system that could take a review from the moving consumer goods sector and determine whether the sentiment expressed was positive or negative.

We went through a process. We started by studying the 3,500-item dataset we would use for training. Then we cleaned the data, turned text into numbers using encoding techniques trained a classifier, tested it thoroughly and looked closely at the types of mistakes it made. Finally we put the working solution into a web application.The model reached an accuracy of 91.5 percent. This result meant we met our objective: creating a functioning sentiment classifier.Still the errors the model made taught us a lot. They showed that sentiment isn't always simple. It’s not about spotting positive or negative words. Sometimes people mix their opinions. They use negations like "not good" or phrases that are neutral but still carry meaning. These examples proved how complex human language is. And why sentiment analysis is not as straightforward as it might seem. Seeing these nuances in action was one of the valuable parts of the project.

14. Future Scope

- We can try using transformer-based models like BERT, DistilBERT or RoBERTa. These models understand language better. May handle tricky cases like negation more accurately. That could lead to results.
- Another option is to keep the model structure but change the task from sentence-level classification to aspect-level classification. This way the model could detect both negative sentiments within the same sentence. For example noting that a product is "tasty" but "too expensive."
- We should also test the model on datasets, including ones in other languages. This includes texts written in Hinglish or code-switched content, where English and Hindi mix. Seeing how well the model performs on data will help us understand its limits.
- Additionally we can improve the web app by adding explainability features. For instance showing which words the model links to negative sentiment would make it easier for users to trust the results.

- We could also build a dashboard for the company. This dashboard could show trends in sentiment across categories. Like how opinions about snacks changed over time. It could provide insights into customer preferences. Highlight areas needing attention.

15. References

- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
- Bird, S., Klein, E. and Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.
- Jurafsky, D. and Martin, J. H. (2024). Speech and Language Processing (3rd ed. draft). Stanford University.
- Pang, B. and Lee, L. (2008). Opinion Mining and Sentiment Analysis. Foundations and Trends in Information Retrieval, 2(1-2), 1-135.

- FMCG Product Reviews dataset (3,500 reviews), fmcg_product_reviews_3500.csv. [https://www.kaggle.com/datasets/viratsharma14/fmcg-customer-sentiments]
- Project web application: https://product-sentiment-analysis-ashen.vercel.app

16. GitHub Repository

GitHub Repository Link: (https://github.com/aadarshsundrani-ops/Product-Sentiment-Analysis)

17. Team Contribution

Team Member	PRN	Specific Contribution
Aadarsh Sundarani	**25030422001**	NLP pipeline implementation, model training /evaluation, project coordination, presentation preparation
Khush Jain	**25030422063**	Dataset Finding, subgroup performance audit, Test via various datasets, model card content, report content.

18. Annexure

Table 6 gives the average rating and sentiment mix for each product category, as supporting material for Section 4.

Category	Reviews	% Negative	% Neutral	% Positive
Baby Care	614	40.4	20.7	38.9
Food & Beverages	583	45.6	15.3	39.1
Health & Hygiene	577	45.9	15.4	38.6
Household Care	586	42.7	16.6	40.8
Oral Care	566	39.8	16.8	43.5
Personal Care	574	44.3	13.9	41.8

Table 6: Sentiment mix by product category